Cryptography

Lecture 9

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The Bigger Picture

- Vulnerability Assessment
- Penetration Testing
- **Firewalls and IDS**
- **Malware & SQL**
- **Denial of Service**
- Social Engineering, Phishing
- Threat Modeling
- **Cryptography**
- **Multi-party Computation**
- EU Cybersecurity Law

Flashback!

- What can firewall do ?
- What is SQL injection?
- How do you execute DoS attack?
- How do you address threats ?

What we learn today

- Types of Cryptography
  - Symmetric Key Encryption (SKE)
    - Historical cipher (Caesar, Vigenère Cipher, Enigma)
    - DES
    - AES (Advanced Encryption Standard)
    - and more
  - Public Key Encryption (PKE)
    - RSA
    - ElGamal encryption
    - and more
- Standardization of cryptography schemes
  - AES, EAXprime

Symmetric Key Encryption

Symmetric Key Encryption

- Historical cipher (Caesar, Vigenère Cipher, Enigma)
- DES
- AES (Advanced Encryption Scheme)
- and more
  - One Time Pad -> Information theoretic secure (most safest)
  - ChaCha20-Poly1305 <- Used in modern internet, TLS/HTTPS, VPN, SSH, Mobile network

Historical Cipher

Historical cipher: Caesar Cipher ~1st century BC

Initial Alphabet: ABCDEFGHIJKLMNOPQRSTUVWXYZ
Encrypting Alphabet: XYZABCDEFGHIJKLMNOPQRSTUVW

Encryption Rule = Left-Shift Alphabet KEY times
KEY = 3 (Caesar used 3 as a key but any number $0 < x < 26$ is fine)

Plaintext: CRYPTOGRAPHY IS FUN
Ciphertext: ZOVMQLDOXMEV FP CRK

Historical cipher: Vigenère Cipher 16th century AC

Encryption Rule = Vigenère Table
Key = CAFE (You can choose)

Plaintext: CRYPTOGRAPHY IS FUN
Ciphertext: ERDTVOKVCPMC KS KUP

How to Encrypt:
Lookup the table with
 CRYPTOGRAPHY (initial alphabet)
 CAFECAFECAFE (Key)

 ERDTVOKVCPMC

Initial Alphabet

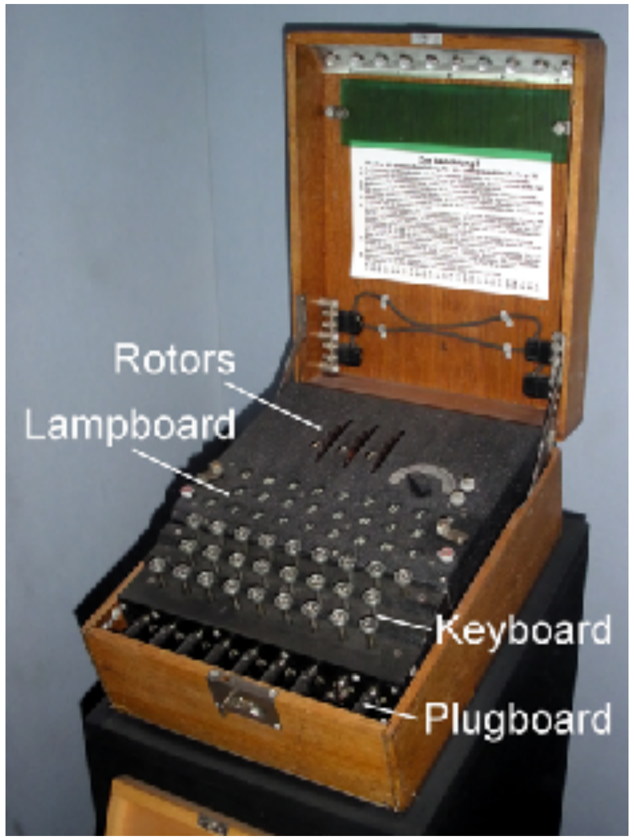
Key

	A	**B**	**C**	**D**	**E**	**F**	**G**	**H**	**I**	**J**	**K**	**L**	**M**	**N**	**O**	**P**	**Q**	**R**	**S**	**T**	**U**	**V**	**W**	**X**	**Y**	**Z**
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

Enigma 1918 ~ 1940s mostly during WWII

Encryption Rule = Use Enigma machine
Key = Combination of Rotors & Plugboard

- Worth 487,000 Euro in 2025 at auction in France
- Movie "The Imitation Game" in 2014 shows Alan Turing who broke the Enigma



Substitution Cipher

- Caesar cipher, Vigenère cipher, and Enigma are so-called substitution ciphers. More specifically,
  - **Monoalphabetic** substitution cipher: Caesar
  - **Polyalphabetic** substitution cipher: Vigenère, Enigma
- The input plaintext (alphabet) is substituted under certain rules
- Complex and well-distributed substitution is necessary

Modern Cryptography

From Classic to Modern cryptography

- Moved from ad-hoc designs to a more scientific approach
- Works with bits and bytes, and not letters/alphabets
- Is developed (to a large degree) in the open

Modern Cryptography

- Intersects with mathematics, theoretical computer science, software engineering and hardware engineering
- Is practiced in academia, industry and intelligence services
- Continuously develops new areas and applications

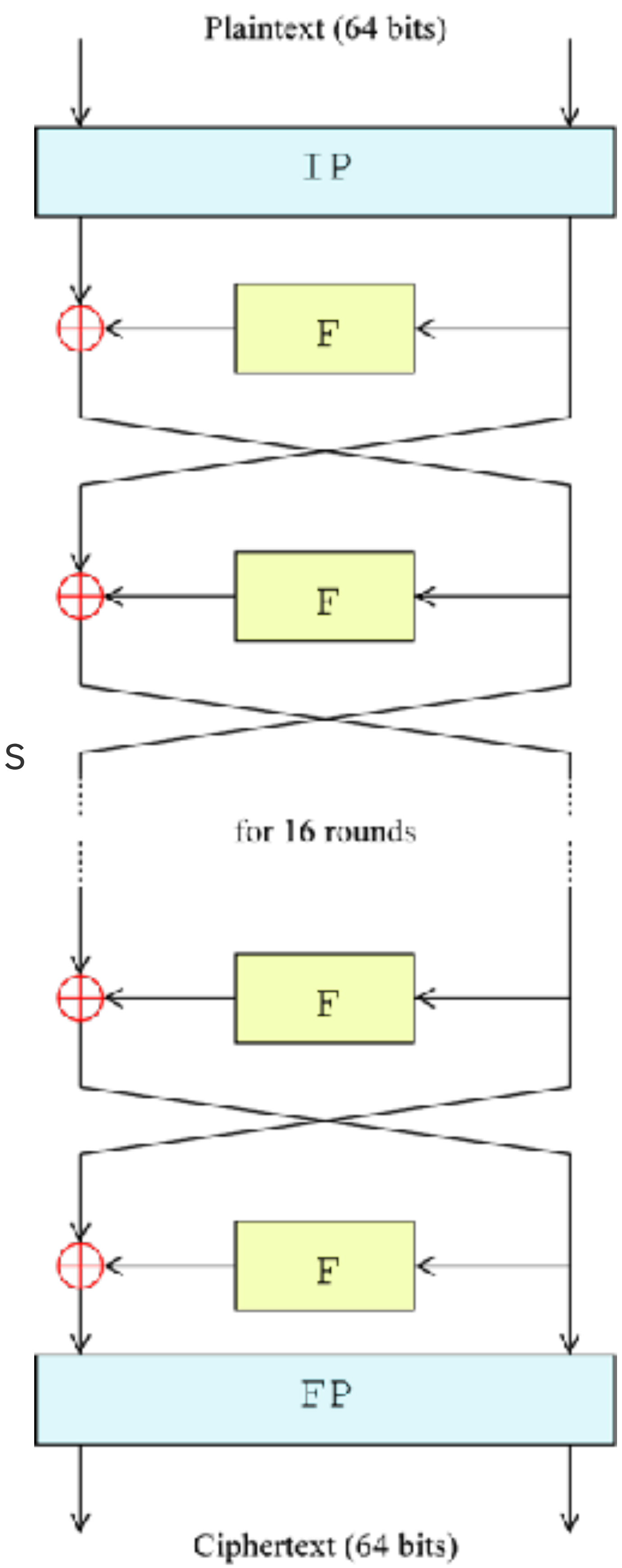
DES

DES (Data Encryption Standard)

- 64 bits input and 64 bits output
- 64 bits key = 56 bits + 8 parity bits
- 16 rounds
- F = Feistel function

  = Substitution +Permutation + other operations

  -> which mixes the dataf
- Short key length allows an attack



DES is not secure anymore

- Late1980s: Differential cryptanalysis is re-discovered by Biham and Shamir
- IBM and NSA knew the attack already, and kept it secret
- 1994: Linear cryptanalysis (more efficient attack) by Mitsuru Matsui
- 2000: variant of linear cryptanalysis with smaller data by **Knudsen** and Mathiassen

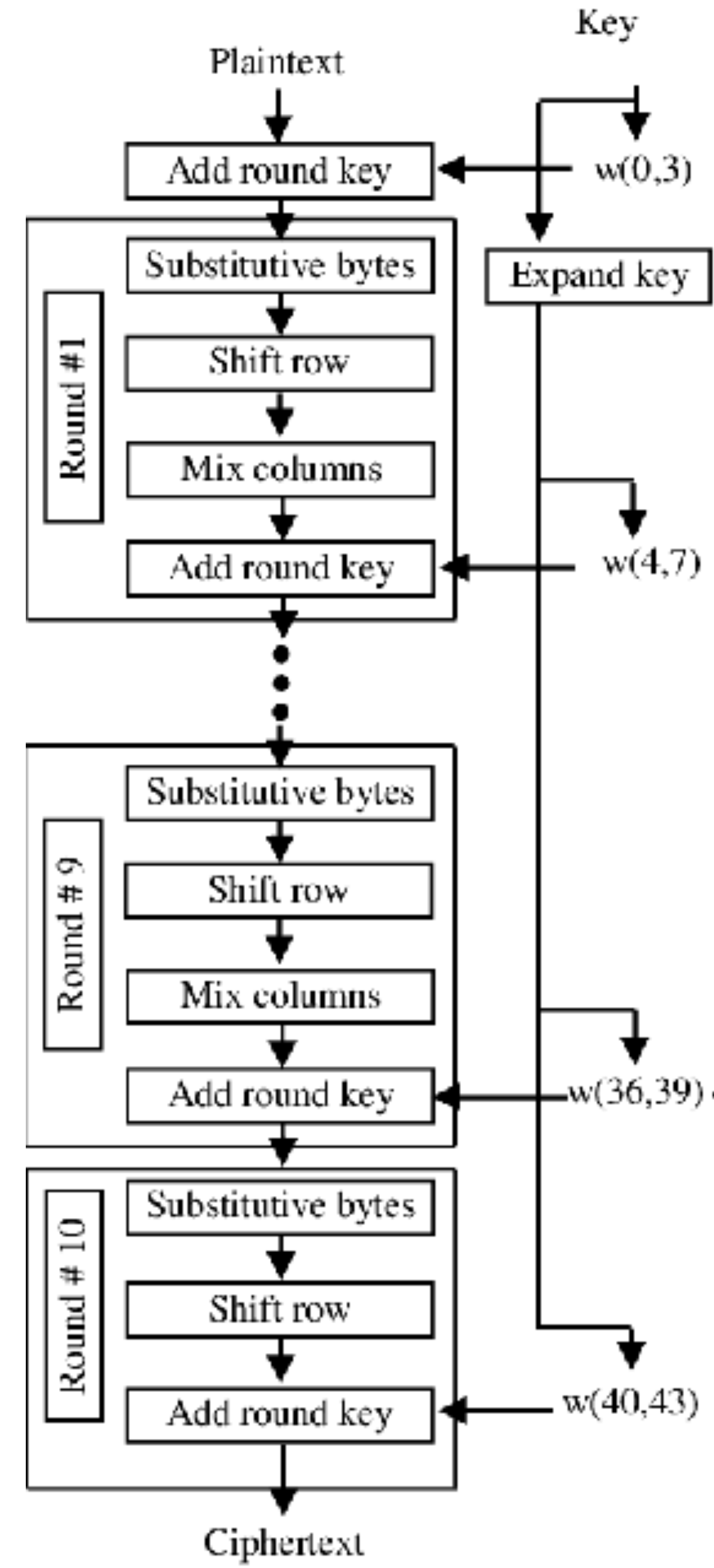
Prof. **Lars Ramkilde Knudsen** at SDU

Providing “Cryptology” course for Msc

AES

AES (Advanced Encryption Standard)

- 128 bit input and 128 bit output
- 128 bit key (option: 192, 256 bits)
- Original name is Rijndael, invented by Rijmen and Daemen
- 10 rounds
- Constructed to avoid differential cryptanalysis and linear cryptanalysis
- Substitution is well-thoughtout to be distributed randomly



AES is not broken…yet

- Key length is 128 bits (nicely long) and substitution operations make the data mixed well, which make the protocol stronger
- No attacks are known
- You can become famous if you find an attack on AES

OTP

One Time Pad 1882~

- Perfect secrecy = Information theoretic secure
- One of the most secure protocols when the key is randomly generated
- The key is equal or longer than plaintext

- Example of bit sequence

```
Plaintext:  1001011…0001011
Key:        1100010…0101110
———————————————————————————(xor)
Ciphertext: 0101001…0100101
```

One Time Pad

- The key must not be reused
- This OTP was used for the (Telegram) hotline between Moscow and Washington, D.C. in 1960s
- It is hard to share the same key
- Once 2 people have the same key, OTP is the most secure way to communicate
- The OTP's idea is used in multi-party computation we learn next week

We learnt: Symmetric Key Encryption

- Historical cipher (Caesar, Vigenère Cipher, Enigma)
- DES
- AES (Advanced Encryption Scheme)
- and more
  - One Time Pad -> Information theoretic secure (most safest)
  - ChaCha20-Poly1305 <- Used in modern internet, TLS/HTTPS, VPN, SSH, Mobile network

Public Key Encryption

Public Key Encryption

- RSA
- ElGamal encryption
- and more (we don’t study today)
  - Digital Signatures
  - Elliptic Curve Encryption
  - Post-quantum cryptography

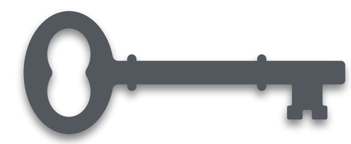
What is PKE?

Encryptor

Public key

Decryptor

Secret Key



- Public key is published and can be used by anyone
- Once the paddle lock is locked, only the person who owns the secret key can unlock it

RSA

RSA

- Rivest, Shamir, and Adleman invented this encryption/decryption scheme in 1977
- They won the ACM Turing Award in 2002 for their invention
- Shamir is also known as the inventor of Shamir Secret Sharing scheme in multi-party computations we learn next week

Textbook RSA with small primes: Overview

Encryptor

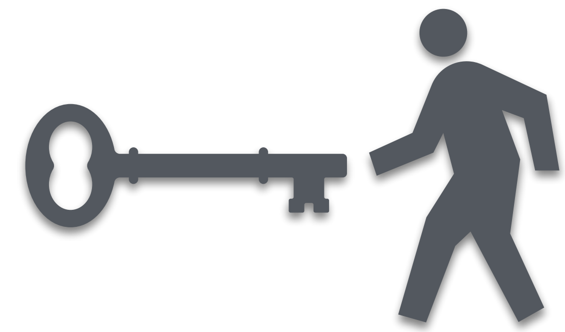
Plaintext

m = 9

⑥

Public Key: (n, e) = (143, 7)

Decryptor



① p=11, q=13

② n = pq = 143

③ φ(n)=(p−1)(q−1)=10×12=120

④ Choose e = 7

⑤ Obtain d = 103

where ed≡1(mod120)

⑦ Encrypt by c = m^e mod n

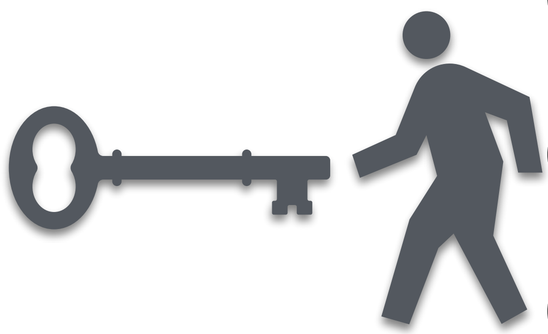
Send c

⑧ Decrypt by m = c^d mod n

Once ①-⑤ is done, execute ⑦⑧ as many as you want

Textbook RSA with small primes: Initial setting

Decryptor



① Pick two prime numbers: $p=11$, $q=13$

② Compute $n = pq = 143$

③ Compute Euler function:

$$\varphi(n)=(p-1)(q-1)=10\times 12=120$$

④ Pick public-value $e = 7$ under a certain rule

⑤ Obtain $d = 103$, where $ed \equiv 1 \pmod{120}$

⑥

Public Key: $(n, e) = (143, 7)$

Secret Key: $(n, d) = (143, 103)$

Textbook RSA with small primes: Encryption

Encryptor **Public Key: (n, e) = (143, 7)**

Plaintext
m = 9

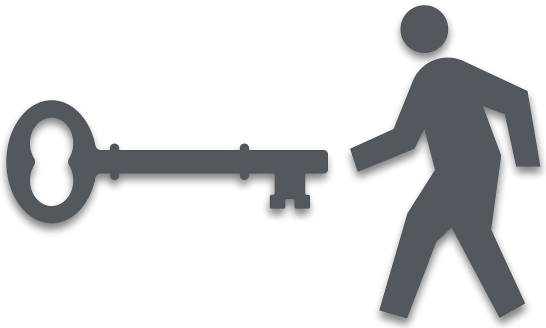
⑦ Encrypt by $c = m^e \bmod n = 9^7 \bmod 143$

$= 48$

This is the ciphertext

Textbook RSA with small primes: Decryption

Decryptor



Public Key: (n, e) = (143, 7)

Secret Key: (n, d) = (143, 103)

⑧ Decrypt by m = c^d mod n = 48^{103} mod 143

= 9

Decrypted!

Notice that we use the property
ed≡1(mod120)

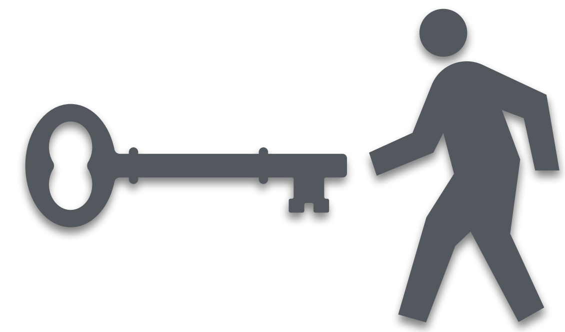
Textbook RSA with small primes: Overview

Encryptor

⑥

Public Key: (n, e) = (143, 7)

Decryptor



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Plaintext

m = 9

⑦ Encrypt by c = m^e mod n

$\xrightarrow{\text{Send c}}$

⑧ Decrypt by m = c^d mod n

Once ①-⑤ is done, execute ⑦⑧ as many as you want

Real World RSA

- Primes must be large and it must be hard to factorize n
- Security of RSA is somewhat related to how difficult it is to factorize n (= pq)
- Key length is 2048
- Textbook RSA is not secure even being constructed as above. Additional smart idea must be combined such as padding, usage of hash function etc.

ElGamal

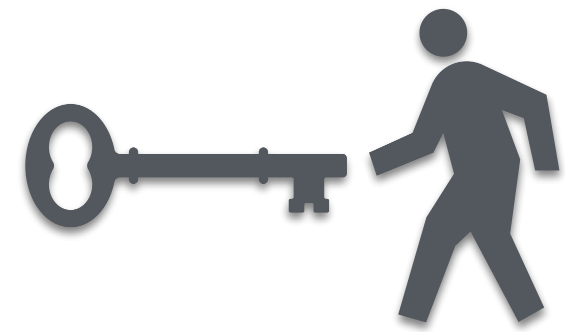
Textbook ElGamal Encryption: Overview

Encryptor

⑤

Public Key: (p, g, h = (23, 5, 8)

Decryptor



Plaintext

m = 9

① p=23

② g = 5

③ Pick secret key x = 6

④ Compute $h = g^x$ mod p

⑥ Pick k = 7

⑦ Encrypt by $c_1 = g^k \bmod p$

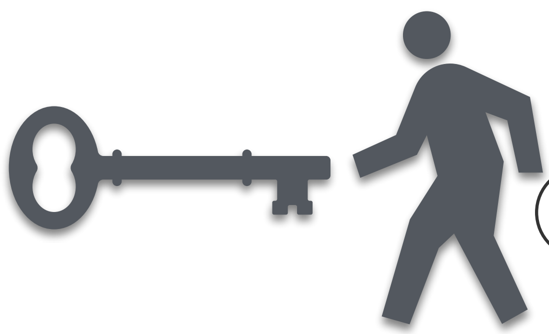
$c_2 = m \cdot h^k \bmod p$

Send (c_1, c_2) ⑧ Decrypt by $s = c_1^x \bmod p$

and $m = c_2 \cdot s^{-1} \bmod p$

Textbook ElGamal: Initial setting

Decryptor



① Pick a prime p=23

② Choose the generator g = 5 of $\mathbb{Z}_p^*$

③ Pick secret key x = 6

④ Compute $h = g^x \bmod p = 5^6 \bmod 23 = 8$

⑤

Public Key: (p, g, h) = (23, 5, 8)

Secret Key: x = 6

Textbook ElGamal: Encryption

Encryptor

Public Key: (p, g, h) = (23, 5, 8)

Plaintext

m = 9

⑥ Pick randomness k = 7

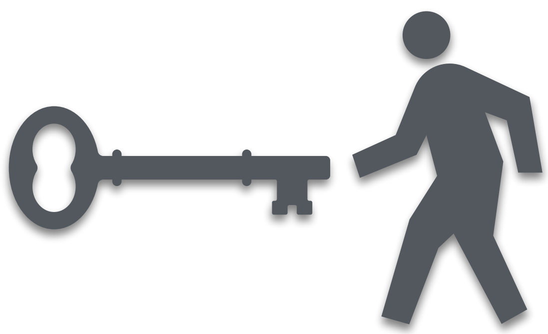
⑦ Encrypt by $c_1 = g^k \bmod p = 5^7 \bmod 23 = 17$

$$c_2 = m \cdot h^k \bmod p = 9 \cdot 8^7 \bmod 23 = 16$$

(c_1, c_2) is the ciphertext

Textbook ElGamal: Decryption

Decryptor



Public Key: (p, g, h) = (23, 5, 8)

Secret Key: x = 6

⑧ Decrypt by $s = c_1^x \bmod p = 17^6 \bmod 23 = 12$

and $m = c_2 \cdot s^{-1} \bmod p = 16 \cdot 2 \bmod 23 = 9$

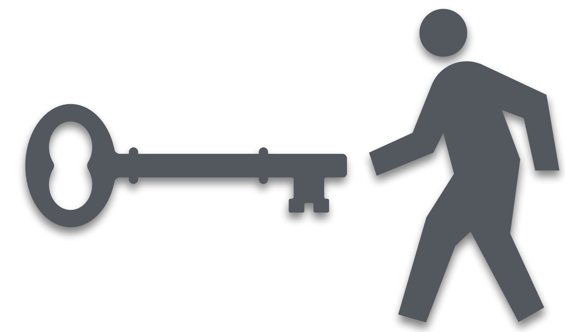
Decrypted!

Notice that we use the fact $12^{-1} \equiv 2 \pmod{23}$

Textbook ElGamal Encryption: Overview

Encryptor

Decryptor



⑤

Public Key: (p, g, h = (23, 5, 8)

Plaintext

m = 9

① p=23

② g = 5

③ Pick secret key x = 6

④ Compute $h = g^x$ mod p

⑥ Pick k = 7

⑦ Encrypt by $c_1 = g^k \bmod p$

$c_2 = m \cdot h^k \bmod p$

Send (c_1, c_2) $\longrightarrow$

⑧ Decrypt by $s = c_1^x \bmod p$

and $m = c_2 \cdot s^{-1} \bmod p$

Standardization

Standardization

- Popular encryption schemes are standardized, ex., FIPS by NIST, ANSI, …
- AES is standardized (FIPS-approved) in 2002 under FIPS PUB 197. During the process…
  - 15 designs were submitted
  - 5 finalists were selected
  - Runner-up design is **Serpent** created by Anderson, Biham, and **Knudsen**

It is not always perfect

- EAX' (EAXprime) was standardized in ANSI C12.22 for transport of electricity meter-based data over a network
- In 2012, Minematsu, Lucks, Morita, and Iwata broke the EAX' and warned the usage of the vulnerable protocol
- Iwata is famous for CMAC which is standardized in NIST SP 800-38B in 2005
- CMAC is used in lower-power devices, wireless protocols like WiFi and Bluetooth

Summary

Summary–Cryptography

- Cryptography works on data level
- Secure transmission over insecure channels
- Can be used to ensure confidentiality & integrity

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  - Public Key Encryption (PKE)
    - RSA
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- Standardization of cryptography schemes
  - AES, EAXprime